# Mechanistic Hypothesis Search — MYH7-Related Skeletal Myopathy

## Hypothesis under evaluation

- **Hypothesis ID:** `shared_myosin_rod_mechanism`
- **Label:** MYH7 and MYH2 light meromyosin (LMM) variants share a common head-sequestration defect
- **Disease:** MYH7-Related Skeletal Myopathy (Mendelian)
- **Status in KB:** EMERGING
- **Seed claim:** Common LMM-region variants in both *MYH7* (slow, type 1 fibres) and *MYH2* (fast, type 2A fibres) produce the *same* loss of the myosin super-relaxed state (SRX) and the *same* raised basal ATP demand, implying that the "rod myosinopathies" share one mechanistic module across fibre types — and, by corollary, that SRX-stabilising modulators tested in one should work in the other.

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## Summary

This report evaluates whether dominant light-meromyosin (rod) variants in *MYH7* and *MYH2* converge on a single, shared disease mechanism: loss of the myosin super-relaxed (SRX) state through impaired head sequestration, with raised basal ATP demand. Over five iterations the investigation combined focused literature search (28 papers reviewed), an original computational sequence/charge analysis of the LMM regions of both genes, and verification queries against GenCC, ClinVar, and ClinicalTrials.gov.

The verdict is **PARTIALLY SUPPORTED (appropriately EMERGING).** The hypothesis rests on one genuinely direct primary study — Carrington et al. 2023 (PMID: 37788100) — that measured, side by side for *MYH7* and *MYH2* LMM variants, the two defining features of the seed claim (reduced SRX fraction and increased basal myosin ATP consumption in patient muscle, with contractility spared). An independent sequence analysis performed here strengthens the structural plausibility: all nine literature-curated *MYH7* LMM pathogenic

residues are strictly conserved in *MYH2*, and the variants are statistically enriched for loss of conserved coiled-coil surface charges (one-sided binomial $p = 1.0 \times 10^{-4}$). So the structural substrate for a shared mechanism is real.

However, the cross-fibre-type generalisation hangs on a **single study with no independent MYH2 replication**, and the proposed molecular bridge — that lost SRX equals impaired folded-back head sequestration (the interacting-heads motif, IHM) — is **actively contested** in the current literature (P 38103642 P 31540109 P 37131793 P 40727670). Competing aggregation and fibre-type loss-of-function mechanisms explain histopathology the energetic model does not, and the therapeutic corollary is untested: ClinicalTrials.gov (2026-10-16) shows **zero SRX-modulator trials** in either skeletal myosinopathy. The hypothesis is a plausible, well-motivated shared-mechanism module for dominant LMM variants that needs independent replication and direct structural confirmation before it can be considered established. It should remain EMERGING.

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## Executive Judgment

**Verdict: PARTIALLY SUPPORTED (appropriately EMERGING).**

The strongest pillar is **Carrington et al. 2023 (PMID: 37788100)**, which analysed common *MYH7* and *MYH2* LMM-region mutations **side by side** using purified LMM proteins, in silico modelling, patient muscle biopsies, mantATP-chase SRX measurements, and X-ray diffraction. It reported the two defining features of the seed claim — decreased SRX fraction and increased basal myosin ATP consumption in vivo, with contractility spared — for variants in **both** genes. The computational analysis in this investigation reinforces the plausibility: all nine curated *MYH7* LMM pathogenic residues are strictly conserved in *MYH2*, and the lesions are enriched for loss of conserved coiled-coil charges (binomial $p = 1.0 \times 10^{-4}$). The structural substrate for a shared mechanism is therefore real and testable.

Three considerations keep the verdict short of "supported." First, the functional-convergence claim depends on one laboratory's one study; there is no independent MYH2 SRX replication. Second, the hypothesis equates "lost SRX" with "impaired folded-back head sequestration (IHM)," and that equivalence is disputed on structural grounds (cardiac vs smooth-muscle IHMs "dramatically different," PMID: 37131793), on X-ray grounds (IHM does not explain relaxed-muscle diffraction, PMID: 31540109), and on assay grounds (mantATP displacement could not robustly quantify an SRX population in porcine cardiac HMM, PMID: 38103642). Third, competing aggregation and fibre-type loss-of-function mechanisms explain much of the histopathology (hyaline bodies, nemaline rods, type 2A fibre loss, rimmed vacuoles) that the energetic/SRX model does not address, and the therapeutic corollary is untested.

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## Key Findings

### Finding 1 — A single direct study underpins the shared SRX-loss / ATP-overconsumption mechanism

Carrington et al. 2023 (PMID: 37788100) is the only study that places *MYH7* and *MYH2* LMM variants in the same experimental frame. Its own words establish the side-by-side design — *"we used multiple approaches to analyze the effects of common MYH7 and MYH2 mutations in the light meromyosin (LMM) region of myosin"* — and the core phenotype — *"basal myosin ATP consumption was increased and the myosin super-relaxed state was decreased in vivo."* The mechanistic interpretation is explicitly head sequestration: *"formation of shutdown heads is impaired, thus increasing myosin head ATP demand in the filaments, rather than affecting contractility."*

The convergent evidence streams were: (1) expressed/purified mutant LMM proteins showing disrupted coiled-coil structure; (2) in-vitro minifilament packing defects; (3) in silico modelling; (4) mantATP chase in patient biopsies showing a reduced SRX fraction; (5) X-ray diffraction showing reduced myosin head order; and (6) evidence of increased basal ATP consumption in vivo. Critically, **myofiber contractility was not affected**, which distinguishes this energetic/regulatory mechanism from a classical force-generation defect and actually argues against a contractile alternative for these rod variants.

The limitation is equally clear: the abstract yields **no per-variant effect sizes, no confidence intervals, and no independent replication.** The effect *direction* is consistent across both genes, but the magnitude and robustness of the cross-fibre-type claim cannot be assessed from the primary report alone, and no second group has reproduced the MYH2 SRX measurement.

## Finding 2 — The structural substrate is demonstrably shared (independent computational confirmation)

An original sequence analysis (UniProt P12883 for MYH7; Q9UKX2 for MYH2; retrieved 2026-10-16) tested whether MYH7 LMM pathogenic variants hit residues that are also present and conserved in MYH2 — a necessary condition for a shared rod mechanism.

**Charge-loss enrichment.** Of 9 literature-curated MYH7 LMM variants, 7 eliminate a positively charged R/K residue. Against a background R/K frequency of 0.177 in the MYH7 LMM region, this is a strong enrichment (one-sided binomial $p = 1.0 \times 10^{-4}$). The remaining lesions are Ala→Pro helix-breakers (A1603P, A1437P). Across the set: 4/9 proline substitutions, 2/9 single-residue deletions, 3/9 charge→aromatic (R→W). **Every variant falls into a canonical coiled-coil-destabilising lesion class.**

**Cross-gene conservation.** A Needleman–Wunsch alignment of the LMM regions gives 526/661 = 79.6% identity, and **all 9 MYH7-mutated residues are strictly identical in MYH2 at aligned positions** (e.g., R1500↔R1506, A1603↔A1609, R1845↔R1851). A heptad a/d hydrophobic-core enrichment test was null (p = 0.32), consistent with the myosin rod being governed by **surface charge periodicity** (the electrostatic "zones" of the coiled-coil zipper) rather than by the hydrophobic core — precisely where charge-removing mutations would be expected to bite.

This is the strongest *parsimony* argument for the hypothesis: if the same residues exist and are conserved in both proteins, and the pathogenic lesions destroy the same class of structural feature, a shared assembly/regulatory consequence is a reasonable default expectation. The caveat (Finding, P 19854198) is that conservation is necessary but not sufficient — identical sites can yield distinct outcomes.

## Finding 3 — Head sequestration (IHM) is structurally established for MYH7, but the IHM↔SRX bridge and cross-myosin universality are contested

Grinzato et al. 2023 (PMID: 37131793) solved the human β-cardiac myosin (MYH7) folded-back **interacting-heads motif (IHM)** at 3.6 Å, confirming that MYH7 heads can sequester into an energy-saving off-state and that off-state interfaces are HCM mutation hot spots — *"The off-state is equated with a folded-back structure known as the interacting head motif (IHM), which is a regulatory feature of all muscle myosins."* This is molecular proof of principle for head sequestration in the slow/β-cardiac isoform.

But the **same study** qualifies universality: *"the structures of cardiac and smooth muscle myosin IHMs are dramatically different. This challenges the concept that the IHM structure is conserved in all muscle types."* That directly complicates the leap from "MYH7 has an IHM" to "all rod myosinopathies share one head-sequestration mechanism across fibre types."

Independently, the equivalence the seed hypothesis relies on — lost SRX = impaired shutdown-head formation — is disputed:

- Mohran et al. 2024 (PMID: 38103642): *"the commonly employed mantATP displacement assay was unable to quantify the population of the SRX state with all data fitting very well by a single exponential"* — a measurement/reproducibility problem for the very readout used in Carrington et al.
- Knupp & Squire 2019 (PMID: 31540109): the IHM does not explain relaxed-muscle X-ray diffraction patterns; there is more than one ordered head state in resting muscle.
- Morel 2025 review (PMID: 40727670): IHM and SRX arise under very different conditions and are weakly or not coupled.

The mechanistic bridge is therefore **biologically plausible but not settled** — this is the single weakest node in the causal chain.

## Finding 4 — Database verification: both genes are bona fide Mendelian myopathy entities, but there are zero SRX-modulator trials

Database checks performed 2026-10-16:

- **GenCC** (submissions export, 30,345 rows): 32 MYH7/MYH2 entries. "MYH7-related skeletal myopathy" is curated **DEFINITIVE** (ClinGen, autosomal dominant) and Strong (Ambry, Labcorp). MYH7 myosin storage myopathy is Moderate–Strong (both AD and AR). MYH2 "myopathy, proximal, and ophthalmoplegia" is **DEFINITIVE** (ClinGen, autosomal recessive), Moderate (ClinGen AD), and Strong/Semidominant (Ambry, Labcorp, PanelApp). Both genes

are therefore established skeletal-myopathy genes with distinct dominant and recessive allelic series.

- **ClinVar** (NCBI E-utilities): MYH7 = 6,011 variant records; MYH2 = 1,795.
- **ClinicalTrials.gov** (API v2): "MYH2 myopathy" = 0 studies; "mavacamten myopathy" = 0; "super-relaxed state myopathy" = 0; "MYH7 distal myopathy" = 1 — but the single hit (NCT04572893) is danicamtiv in dilated cardiomyopathy, **not** a skeletal myopathy. This is a **verified null** for SRX-modulator trials in either skeletal myosinopathy.

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## Mechanistic Model / Interpretation

The hypothesis proposes the following causal chain. Strength of evidence is annotated at each link:

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LMM-region point mutation (MYH7 or MYH2)
      │   [STRONG — variants catalogued; charge/proline lesion classes; this report F2]
      ▼
Loss of conserved coiled-coil surface charge / helix-breaking
      │   [STRONG in vitro — <a href="https://pubmed.ncbi.nlm.nih.gov/29660325/" rel="noopener
      ▼
Destabilised rod → aberrant thick-filament packing / assembly
      │   [STRONG in vitro — minifilament EM defects, <a href="https://pubmed.ncbi.nlm.nih.gov
      ▼
Impaired folded-back "shutdown head" (IHM) formation
      │   [CONTESTED — IHM real for MYH7 (<a href="https://pubmed.ncbi.nlm.nih.gov/37131793/"
      │    equivalence disputed (<a href="https://pubmed.ncbi.nlm.nih.gov/38103642/" rel="noop
      ▼
Loss of super-relaxed (SRX) state → raised basal myosin ATP demand
      │   [SINGLE STUDY, both genes — <a href="https://pubmed.ncbi.nlm.nih.gov/37788100/" rel=
      │    questioned <a href="https://pubmed.ncbi.nlm.nih.gov/38103642/" rel="noopener norefe
      ▼
Energetic stress across fibre types (type 1 MYH7 / type 2A MYH2)
      │   [INFERRED — not independently replicated; contractility spared]
      ▼
Clinical myopathy (distal/proximal weakness, ophthalmoplegia for MYH2)
          [Clinical entities ESTABLISHED, but causal link to SRX loss INFERRED]
```

The **upstream half** of this chain (mutation → coiled-coil destabilisation → filament packing defect) is well supported by multiple in-vitro studies and by the conservation/charge analysis in this report. The **downstream half** (packing defect → IHM loss → SRX loss → energetic stress → clinical phenotype) rests critically on a single study for the SRX/ATP readout, and the IHM→SRX intermediary node is the weakest link in the entire chain.

An important nuance the energetic model does **not** cover: the characteristic histopathology differs by gene. MYH7 rod/storage myopathies show **hyaline bodies** in type 1 fibres (P 15699387 P 17336526 ); P 25666907 MYH2 myopathy shows **type 2A fibre loss, rimmed vacuoles, and even nemaline rods/ filamentous tangles** (P 33926564 P 32578970 ). P 29934118 A purely energetic SRX-loss mechanism predicts fibre-type energetic stress but not these distinct aggregation/structural pathologies, implying that **additional, gene- and fibre-type-specific mechanisms operate alongside any shared energetic module.**

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# Evidence Matrix

| Citation | Evidence type | Role | Mechanistic claim tested | Key finding | Subtype/ context | Cor & lim |
|---|---|---|---|---|---|---|
| 37788100 Carrington 2023 | In vitro + human biopsy | **SUPPORT (primary)** | MYH7 & MYH2 LMM variants both lose SRX, raise basal ATP, spare contractility | SRX ↓, basal ATP ↑ in vivo; shutdown-head formation impaired | Both genes, LMM region | Dir side side sing stu per stat rep |
| This report (F2) | Computational | **SUPPORT (parsimony)** | Shared structural substrate | 9/9 MYH7 LMM sites conserved in MYH2; charge-loss enriched p=1e-4 | LMM coiled-coil | Seq leve infe mea sha fun |
| 37131793 Grinzato 2023 | Structural (cryo-EM) | **SUPPORT** + **QUALIFY** | MYH7 head sequestration (IHM) exists; universality | 3.6 Å MYH7 IHM; but cardiac vs smooth IHM "dramatically different" | β-cardiac/MYH7 | Cor IHM MY und cro my uni |
| 38103642 Mohran 2024 | In vitro | **QUALIFY / CHALLENGE** | SRX measurable by mantATP? | mantATP displacement couldn't quantify SRX (single-exponential fits) | Porcine cardiac HMM | Cha the cen the hyp |
| 31540109 Knupp & Squire 2019 | Computational/ structural | **CHALLENGE** | IHM = relaxed-state structure? | IHM does not explain relaxed-muscle X-ray patterns; >1 ordered head state | Fish skeletal, insect | Que IHM equ |
| | Review | | | | | |

| Citation | Evidence type | Role | Mechanistic claim tested | Key finding | Subtype/ context | Cor & lim |
|---|---|---|---|---|---|---|
| 40727670 Morel 2025 | | **CHALLENGE (review-level)** | IHM–SRX coupling | IHM & SRX arise under very different conditions; weak/ absent coupling | Thick filaments, general | Rev orie but on t disp nod |
| 29660325 2018 | In vitro + cell | **SUPPORT (upstream)** | LMM lesions destabilise coiled coil & filament assembly | A1603P, K1617del ↓ helicity, disrupt minifilaments, aberrant M-line packing | MYH7 Laing MPD-1 | Stro ups sup SRX rea |
| 19854198 2009 | In vitro | **QUALIFY** | Same residue, different disease | R1500P (skeletal) vs R1500W (DCM) have *distinct* molecular phenotypes | MYH7 rod | Wa ide site ide out |
| 39216086 2024 | In vitro + mouse | **COMPETING / therapeutic** | Mavacamten rescues SRX energetics | Rescued ATP overconsumption in vitro; failed to fix energy proteome in vivo (4 wk) | NEB nemaline myopathy | Sho targ wor vitr sho in-v effi dou |
| 33926564 2021 | Human clinical | **COMPETING (pathology)** | MYH2 pathology | Nemaline rods + filamentous tangles; myofibril disorganisation | MYH2 (incl. motor-domain variant) | Agg pat not exp by S mo |
| | Human clinical | | | | | |

| Citation | Evidence type | Role | Mechanistic claim tested | Key finding | Subtype/ context | Cor & lim |
|---|---|---|---|---|---|---|
| 36380287 2022 | | **SUPPORT (context)** | Dominant MYH2 tail/ assembly variant | Splice loss of aa1860-1891 (assembly region); type 2A loss | Dominant MYH2 | Lin dist ass reg dor dis |
| 25666907 2015 | Human clinical | **COMPETING (pathology)** | MYH7 storage pathology | R1820W → myosin storage, hyaline bodies in type 1 fibres | Recessive MYH7 MSM | Agg pat bey ene mo |
| 17336526 2007 | Human clinical | **SUPPORT (context)** | Common MYH7 rod variant | R1845W recurrent; phenotypic variability; hyaline bodies in only half | MYH7 MSM/ scapuloperoneal | Esta a "c LM var var exp |

## Data and Tool Use

| Source | Accession / URI | Access status | Query / filter | Relevance |
|---|---|---|---|---|
| PubMed | NCBI E-utilities | **Accessed** | MYH7/MYH2 LMM, SRX, IHM, head sequestration; 28 papers reviewed | Directly relevant |
| UniProt | P12883 (MYH7), Q9UKX2 (MYH2); retrieved 2026-10-16 | **Accessed** | LMM region sequences for alignment & charge analysis | Directly relevant |
| GenCC | submissions export, 30,345 rows; 2026-10-16 | **Accessed** | Filter to MYH7/MYH2 → 32 entries | Directly relevant; gene-validity confirmation |
| ClinVar | NCBI E-utilities; 2026-10-16 | **Accessed** | MYH7 (6,011 records), MYH2 (1,795) | Context; variant counts |
| ClinicalTrials.gov | API v2; 2026-10-16 | **Accessed — verified null** | "MYH2 myopathy"=0; "mavacamten myopathy"=0; "super-relaxed state myopathy"=0; "MYH7 distal myopathy"=1 (NCT04572893, DCM, off-target) | Verified absence of SRX-modulator trials |

**Analyses inventory:**

1. *Sequence conservation & charge analysis* (Finding 2) — Input: UniProt P12883, Q9UKX2. Method: Needleman–Wunsch alignment of LMM regions; one-sided binomial enrichment test for charge loss; heptad a/d core test. Output: 9/9 conserved sites; charge-loss p = 1.0e-4; core test null p = 0.32. **Outcome: succeeded.**
2. *Database verification* (Finding 4) — GenCC/ClinVar/ClinicalTrials.gov queries. **Outcome: succeeded, including a verified negative for SRX-modulator trials.**

3. *Wet-lab / patient-level SRX or ATP analysis* — **Not attempted; no accessible dataset.** All functional SRX evidence is from the literature (primarily
   P 37788100
   ).
   No analysis was silently substituted for a failed one; the functional-convergence claim rests on literature, which is stated explicitly as a limitation.

Artifact bundle location (as specified): `kb/hypotheses/MYH7-Related_Skeletal_Myopathy/shared_myosin_rod_mechanism/openscientist_artifacts` — containing the sequence-analysis code/outputs, the LMM alignment, the database query logs, and the verified-null ClinicalTrials.gov search log per the per-iteration records.

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## Limitations and Knowledge Gaps

1. **Single-study dependence for functional convergence.** The entire cross-fibre-type SRX/ATP claim comes from one report (P 37788100 ).
   *Checked:* PubMed, no independent MYH2 SRX replication found as of 2026-10-16. *Resolution:* an independent lab repeating mantATP-chase SRX and X-ray diffraction on MYH2 patient biopsies.
2. **Contested IHM→SRX bridge.** The hypothesis equates lost SRX with impaired head sequestration, but IHM–SRX coupling is disputed (P 38103642 P 31540109 P 40727670 )
   and the SRX assay itself is questioned. This is the weakest causal node. *Resolution:* direct cryo-EM/X-ray demonstration that LMM-variant filaments have fewer folded-back heads, paired with an orthogonal SRX assay.

3. **Cross-myosin structural universality unproven.** Cardiac and smooth-muscle IHMs differ "dramatically" ( P 37131793 ); no MYH2 IHM structure exists. *Resolution:* solve the MYH2 (type 2A) IHM and compare interfaces to MYH7.
4. **Pathology not explained by the energetic model.** Hyaline bodies (MYH7) vs type 2A loss / rimmed vacuoles / nemaline rods (MYH2) point to additional aggregation/loss-of-function mechanisms. *Resolution:* correlate SRX loss with aggregation burden within the same biopsies.
5. **Same residue ≠ same outcome.** R1500P vs R1500W give distinct molecular phenotypes ( P 19854198 ), warning that conserved sites do not guarantee identical downstream effects. Conservation (Finding 2) is necessary but not sufficient.
6. **Therapeutic corollary untested; in-vivo efficacy doubtful.** Zero SRX-modulator trials exist for either skeletal myopathy (verified null, 2026-10-16), and mavacamten rescued SRX energetics *in vitro* but failed to correct the energy proteome *in vivo* over 4 weeks in a related nemaline model ( P 39216086 ). *Resolution:* proof-of-concept ex-vivo SRX-stabiliser study on MYH2 and MYH7 LMM-variant fibres, then a biomarker-driven trial.

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## Alternative / Competing Models

- **Aggregation / proteostasis toxicity (parallel mechanism).** Misassembled mutant myosin forms hyaline bodies (MYH7) or nemaline rods/tangles (MYH2); a gain-of-function toxicity independent of SRX energetics. Likely operates *alongside* the energetic module. ( P 25666907 ) P 33926564
- **Fibre-type loss-of-function (alternative, especially recessive MYH2).** Recessive MYH2 causes near-complete loss of type 2A fibres — a developmental/structural deficiency, not an

energetic one. A genuine alternative for the recessive allelic series.
( P 32578970 ) P 29934118

- **Filament-assembly haploinsufficiency (upstream cause).** Distal-tail / assembly-competence-domain variants (e.g., MYH2 aa1860-1891 splice loss) impair incorporation into thick filaments — upstream of any SRX consequence.
( P 36380287 )
- **Classical contractile / force defect (refuted for LMM variants).** Carrington et al. explicitly found contractility spared, arguing *against* a force-generation mechanism for these rod variants — a point in favour of the regulatory/energetic model over this alternative.

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## Discriminating Tests

1. **Independent MYH2 SRX replication.** Stratify: dominant MYH2 LMM-variant patients vs controls. Assay: mantATP-chase SRX + X-ray diffraction on biopsies, *plus* an orthogonal SRX readout to control for the assay critique
( P 38103642 ).
Expected if hypothesis true: reduced SRX fraction mirroring MYH7.
2. **Direct IHM quantification on variant filaments.** Cryo-EM of reconstituted MYH7 and MYH2 LMM-variant minifilaments; count folded-back heads. Expected: fewer IHM heads in variants of both genes.
3. **Solve the MYH2 (type 2A) IHM** and compare off-state interfaces to the MYH7 IHM
( P 37131793 )
— tests structural universality directly.
4. **Within-biopsy correlation of SRX loss vs aggregation pathology.** Tests whether the energetic module is independent of, or coupled to, hyaline-body/nemaline-rod formation.
5. **Ex-vivo SRX-stabiliser (mavacamten/analogue) rescue** on MYH2 and MYH7 LMM-variant fibres, with energy-proteome readout, acknowledging the in-vivo efficacy caveat

( ).
P 39216086
Expected if hypothesis true: restored SRX and reduced ATP overconsumption in both.

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## Curation Leads (require curator verification)

- **Keep status EMERGING.** Do not promote; functional convergence is single-study and the IHM→SRX node is contested.
- **Add qualifying evidence edges:**
- P 37131793 — snippet: *"the structures of cardiac and smooth muscle myosin IHMs are dramatically different. This challenges the concept that the IHM structure is conserved in all muscle types."* Role: QUALIFY universality.
- P 38103642 — snippet: *"the commonly employed mantATP displacement assay was unable to quantify the population of the SRX state with all data fitting very well by a single exponential."* Role: CHALLENGE assay/readout.
- P 31540109 — snippet: *"the interacting head motif does not, in fact, explain what is observed."* Role: CHALLENGE IHM↔relaxed-state equivalence.
- P 19854198 — snippet: *"each mutation displays distinct structural, thermodynamic, and functional properties."* Role: QUALIFY (same residue ≠ same outcome).
- **Candidate pathophysiology nodes/edges:** `LMM coiled-coil charge loss → filament-packing defect` (strong); `filament-packing defect → impaired IHM/shutdown-head` (contested — flag); `impaired SRX → raised basal ATP demand` (single-study); `SRX loss → clinical weakness` (inferred).
- **Candidate ontology terms:** type 1 slow-twitch skeletal muscle fibre; type 2A fast-twitch fibre; myosin super-relaxed state (GO: regulation of ATPase activity / striated muscle myosin thick filament assembly); interacting-heads motif.

- **Candidate subtype restriction:** limit the shared-mechanism claim to **dominant LMM/rod variants**; recessive MYH2 (type 2A fibre loss) likely follows a distinct loss-of-function mechanism.
- **Candidate knowledge_gaps entries:** (1) no independent MYH2 SRX replication as of 2026-10-16; (2) IHM↔SRX coupling unresolved; (3) verified null — zero SRX-modulator trials in either skeletal myosinopathy (ClinicalTrials.gov v2, 2026-10-16).

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## Evidence Base Summary

The hypothesis is supported chiefly by one direct primary study (P 37788100) and by an independent conservation/charge analysis (this report). It is reinforced upstream by coiled-coil destabilisation studies (P 29660325) and by structural proof of a MYH7 IHM off-state (P 37131793). It is qualified or challenged by the contested IHM–SRX coupling (P 38103642 P 31540109), P 40727670 by the "same residue, different disease" caution (P 19854198), and by aggregation/fibre-type pathologies that an energetic model does not explain (P 25666907 P 33926564 P 32578970). The therapeutic corollary is unproven and partly undercut by a failed in-vivo mavacamten

energy-proteome rescue ( P 39216086 ) and a verified absence of relevant trials. Overall: a plausible, well-motivated shared-mechanism module for dominant LMM variants that remains appropriately EMERGING pending independent replication and direct structural confirmation.

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